



Development of reduced and optimized mechanism for ammonia/hydrogen mixture based on genetic algorithm

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ARTICLE INFO

Handling Editor: Henrik Lund

Keywords:

Ammonia/hydrogen mixture
Engine simulation
Mechanism reduction and optimization
Genetic algorithm

ABSTRACT

As carbon-free and renewable fuels, ammonia and hydrogen are promising fuels to be utilized in combustion devices. For this, the genetic algorithm was employed to reduce and optimize the mechanism for ammonia/hydrogen mixture, which was aimed to develop a compact and reliable skeletal mechanism for real engine simulation with ammonia as fuel. During the progress of GA reduction, the extracted sub-mechanisms were constrained high fidelity with original mechanism on predicting the combustion characteristics through the definition of penalty functions. Besides, the number of reactions and computer calculation time of the sub-mechanism were simultaneously tracked to reduce the chemical stiffness. The resulting mechanism contained 29 species and 63 reactions and was streamlined to be coupled into complex CFD calculation. The reduced mechanism was adjusted to better capture fundamental combustion data and the mechanism optimization was finished automatically through GA operation. To verify the reliability of the optimized mechanism, the experimental data including ignition delay times, laminar flame speed, JSR species concentration, premixed flame structure and HCCI engine in-cylinder pressure at various conditions was utilized to evaluate the performance of the models. As a result, the optimized mechanism was capable to reproduce those measurements and was credible for engine combustion simulation.

1. Introduction

The combustion of fossil fuels produces greenhouse gases, mainly CO₂, which will accelerate global warming and jeopardize human living space [1]. As carbon-free, renewable and sustainable fuels, the ammonia and hydrogen are considered to be promising to substitute fossil fuels [2, 3]. Ammonia has some problems such as low burning rate and unstable combustion, which will be improved with the mixing of hydrogen [4,5]. What's more, ammonia is regarded as one of the ideal carriers for hydrogen due to its low reactivity. It was reported that 1 m³ of liquid ammonia had the capacity to store 108 kg of H₂ at a temperature of 20 °C and pressure of 8.6 atm [6]. Due to the high hydrogen content in ammonia, more heat could be released during ammonia oxidation. It can be used as a chemical storage medium for hydrogen and also release hydrogen by decomposition [7]. Exploring the feasibility of complementary application of ammonia-hydrogen mixtures in internal combustion engines (ICE), gas turbines and other practical power engines attracted a lot of attention [8,9].

To reduce emissions in the transportation industry currently, the compression ignition (CI) engines and spark ignition (SI) engines fueled

with ammonia and hydrogen mixture was studied to provide stable and reliable power output. Dimitriou et al. [10] reviewed the research on dual-fuel compression ignition engine operation with ammonia and other highly reactive fuels like diesel. Pochet et al. [11] successfully conducted experiments on the ammonia-hydrogen mixture HCCI engines with an effective compression ratio of 22:1 and pointed out that the ammonia-hydrogen blending ratio greatly affected the engine output except for the indicated efficiency. Although it was difficult to ignite ammonia alone, the SI engine worked satisfactorily with low and moderate hydrogen addition [12]. Frigo et al. [13] confirmed the necessity of blending hydrogen for ignition improvement and an increase in combustion velocity when ammonia was applied in gasoline engines, in addition, the appropriate proportion depended on the situation.

The foundation chemistry mechanism of ammonia was laid by Miller et al. [14] and Lindstedt et al. Unlike methane and hydrogen, the clarity of understanding of the chemical kinetic behavior of ammonia remains a great challenge, especially at low temperatures ($T < 1200$ K) and under dilute mixtures. A number of chemical kinetic mechanisms for ammonia, ammonia/hydrocarbon mixture were proposed and verified, e.g. Glarborg et al. [15], Klippenstein et al. [16], Mathieu et al. [17], Me'vel et al. [18], Song et al. [19], Stagni et al. [20], Zhang et al. [21], Otomo et al.

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Abbreviations

GA	genetic algorithm
CFD	computational fluid dynamics
SI	spark ignition
CI	compression ignition
HCCI	homogeneous charge compression ignition
FSSA	full species sensitivity analysis
LUMP	lumped methods
IDTs	ignition delay times
RCM	rapid compression machine
JSR	jet-stirred reactor
LFS	laminar flame speed
DNS	direct numerical simulation
LES	large eddy simulation
DRG	directed relation graph
DRGEP	directed relation graph with error propagation
QSSA	quasi-steady-state assumption
EF	element flux analysis
FP	floating point
Mechanism 1	Reduced mechanism
Mechanism 2	Optimized mechanism

[22], NUI 2017 [23], Nakamura 2017 [24]. Stagni et al. [20] conducted ammonia thinning combustion experiments in a jet stirred and flow reactor covering the temperature range of 500–2000 K. It was found that H₂NO played an important role in ignition delay control, and the reaction $\text{NH}_2 + \text{O} = \text{HNO} + \text{H}$ reduced the predicted laminar flame velocity while promoting NO production at high temperatures. Liao et al. [25] measured the ignition delay times of ammonia added N₂O or H₂ in a RCM and modified the mechanism of ammonia. It was found that the addition of H₂ promoted the production of OH radicals and triggered the first stage of ignition. Otomo et al. [22] improved the mechanism by relevant elementary reactions including NH₂, HNO, and N₂H₂. It was found that reaction $\text{HNO} + \text{OH}$ played the most important role for NO production during lean combustion, and the reaction channel with the largest contribution was $\text{HNO} + \text{H}$ for rich combustion. Zhang et al. [21] investigated ammonia/hydrocarbon mixture oxidation through jet-stirred reactor (JSR) experiments at intermediate temperatures. The study showed the mixing of H₂ enhanced the oxidative reactivity of NH₃, and the oxidation radicals inhibited the formation of N₂O, while the enhanced NO promoted its formation.

The mechanism needs to be reduced while providing a reasonable accuracy to pursue a proper and cost-effective numerical simulation, especially direct numerical simulation (DNS) and large eddy simulation (LES) of a real engine. In addition, numerical simulation for combustion behavior in the dual-fuel engine may face a heavy burden when a detailed ammonia mechanism is combined with the mechanism of another active fuel like dimethyl ether. The reduced or skeletal mechanism of ammonia/hydrogen mixture was relatively limited in literature works. Xiao et al. [26] reduced the scale of the ammonia/hydrogen mechanism for 2D gas turbine LES. The mechanism reduction was performed in Chemkin II software using the directed relation graph (DRG) method. The reduced mechanism developed by Duynslaegher et al. [27] for low-pressure conditions was verified only under species concentration. Rocha et al. [28] considered that the current ammonia/hydrogen combustion mechanism had the potential for further simplification with high fidelity. Three reduced mechanisms with 50–70 element reactions were proposed based on the mechanism of Mathieu [17]'s, Okafor [29]'s and Otomo [22]'s. Li et al. [30] concentrated on extracting and reducing the NH₃ and NH₃/H₂ mechanism from the large-scale mechanism with directed relation graph with error propagation (DRGEP) and DRGEP with sensitivity analysis (DRGEP-SA) strategy. Still, 28 species and 213

reactions were included in his NH₃/H₂ combustion mechanism. Cai et al. [31] developed reduced NH₃/H₂ mechanisms for ignition delay and laminar flame speed prediction with Chemkin-Pro 2019 using a method of direct relation graph with error propagation. The calculated IDTs by his mechanism was compared with experimental data only under high temperature and pure ammonia fueled. From the above papers, compact kinetic mechanism of ammonia/hydrogen required more extensive validation against not only original mechanism but also experimental data. Especially, the evaluation of the ignition delay times (IDTs) at both low- and high-temperature should be considered, which is partially regarded as ICE simulations of practical interest [32]. In addition, the skeletal mechanism with the small number of reaction rate parameters can be optimized to achieve a better mechanism configuration.

Attractive approaches to the reduced mechanisms such as the decoupling method, DRG, quasi-steady-state assumption (QSSA), element flux analysis (EF), and sensitivity analysis all require the user to understand the detailed chemistry knowledge. The method of uncertainty quantification was used for mechanism optimization in order to minimize the deviation between calculation and measurement in Wang [33]'s study. The mechanism optimization by manual iteration strongly depends on path analysis and experience [34]. Therefore, researchers introduce genetic algorithms (GA) to develop a simple and robust reduction methodology. The optimal solution search process is directed by the overall objective function, and unimportant reactions would be automatically removed in the iterative algorithm. The penalty function is used to constrain CPU calculation time, avoiding stiff and slow converging mechanism set. The method was proved to be effective in previous studies [35–37]. GA can also be exploited in optimizing the reaction rate constants to decrease the gap between model predictions and experimental measurements [32,34,38]. In addition, a traditional algorithm for finding an optimal solution like gradient-based algorithms and the solution mapping method was unsuitable for mechanism tuning due to the complexity and non-linearity of the chemical mechanism [39]. Chang et al. [34] optimized the skeletal oxidation mechanism of n-pentanol by tuning the Arrhenius pre-exponential factors of C₄-C_n sub-mechanism using a multi-objective genetic algorithm. As a result, predictions by the optimized mechanism model were well agreed with the experimental IDTs, laminar flame speed and species concentrations in JSRs.

In this study, the ammonia/hydrogen reduced mechanism was developed by genetic algorithm-based method. Penalty functions were designed to guide the removal of uninterested reactions, in which not only the fidelity between the streamlined mechanism and the complete mechanism was considered but also the CPU computational time was limited to reduce the stiffness of the mechanism set. The reduced mechanism for ammonia/hydrogen mixture was also optimized based on GA for better prediction of the ignition delay times and laminar flame speed. Finally, the mechanisms were validated by contrast with experimental data from fundamental combustion reactors and in-cylinder pressure profiles from the HCCI engine. The main content of this article was presented in Fig. 1.

The contribution of this paper to the related field is mainly twofold. First, the genetic algorithm, a simple yet robust method, was tested for reducing and optimizing ammonia/hydrogen mechanism, in which the role of some elementary reactions still was unknown currently. Second, not only developing a compact ammonia/hydrogen simplified mechanism for engine CFD simulation, but also performing mechanism optimization to achieve a better mechanism configuration. The extensive validation proved confidence in the reliability of the reduced mechanism in this paper.

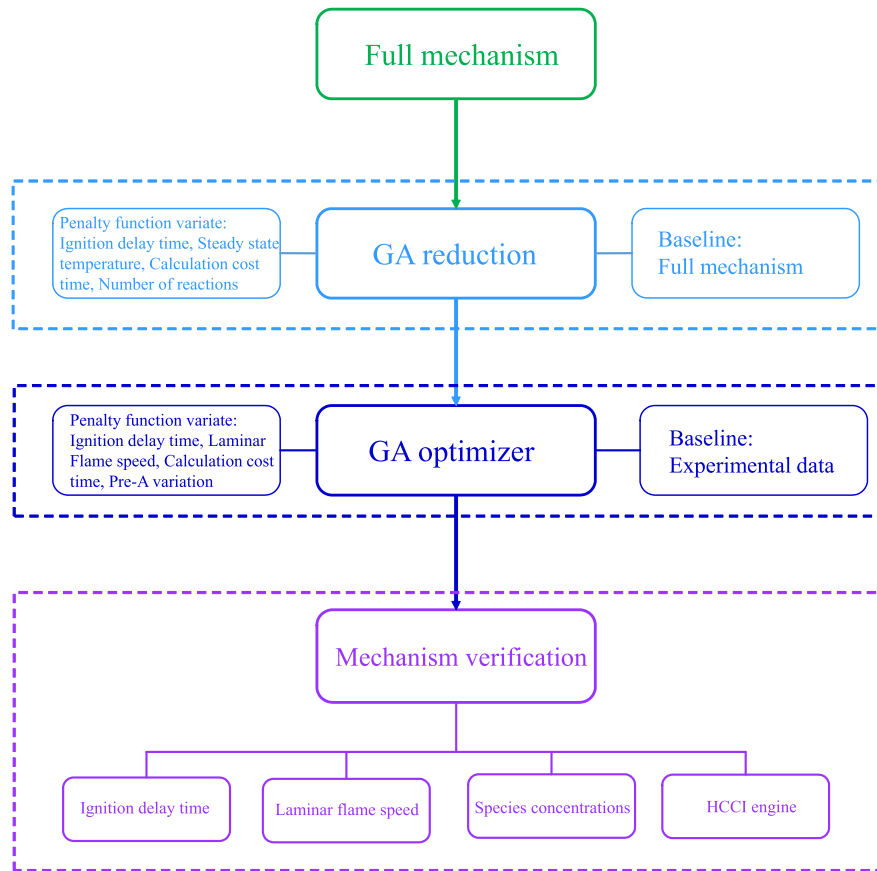


Fig. 1. The flow chart of the research method of this article.

2. Methodology

2.1. Genetic algorithm

Genetic algorithm is an adaptive heuristic mathematical technique based on biological evolution principles. A set of initial populations are created firstly in the solution space of an optimal problem in GA. Then the behavior of each individual is evaluated by the fitness function, which correlates with the optimization problem and therefore guides genetic operators so that individuals in the next generation of populations behave better. The iterative process between generations is also the search process of the optimal solution, and the process terminates when an established criterion is met or a given number of generations are gained [35]. The advantage of GA includes fast and stochastic search capability, potential parallelism, and search using evaluation function heuristics.

The genetic algorithm procedure adopted is shown in Fig. 2 and the search process for the optimal solution in this study consists of the following steps:

1. Initialization, where a set of initial individuals (single chromosome in one individual) with reaction information are generated.
2. Evaluation, where the performance of individual is evaluated and corresponding fitness is assigned to each individual.
3. Selection, where the chromosome of the fittest individual is retained for next-generation (Elitist). The other individuals compete for the opportunity to retain chromosomes for the next generation.
4. Crossover, where the chosen chromosomes exchange information to create a new mechanism set for the next generation.

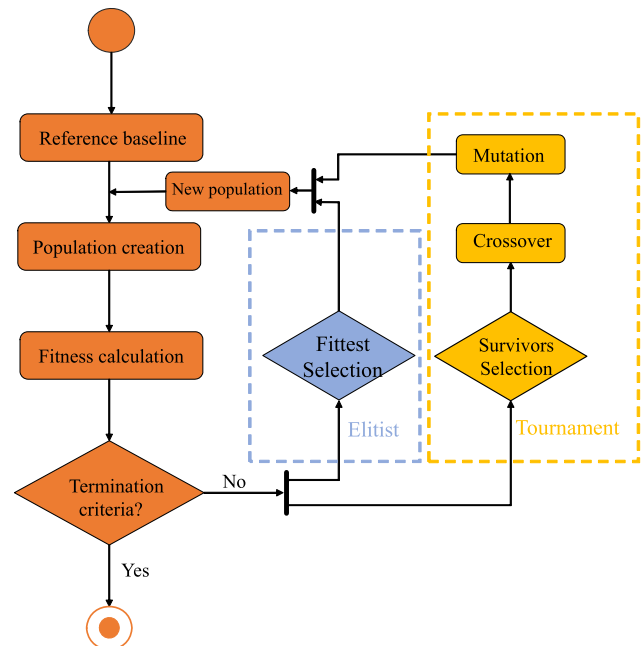


Fig. 2. Activity chart of the genetic algorithm.

5. Mutation, where a small number of the children's genes are changed. For mechanism reduction, this gene operation corresponds to the deletion of chemical reaction, and for mechanism optimization, it means the modification of the reaction rate coefficient.

2.2. Implementation details of mechanism reduction

The kinetic mechanism reduction method based on genetic algorithms developed by Sikalo et al. [37] is adopted in this study, and the same operation has appeared in the literature [35,36]. Therefore, a brief description of the implementation method is presented here. The GA is used to eliminate reactions instead of species. It is possible to achieve a more streamlined structure by removing unimportant reaction paths compared with deleting the species. Therefore, each gene corresponds to one reaction and the gene takes the value 0 (meaning reaction absent) or 1 (meaning reaction presence). The process of chromosome encoding and decoding can also be ignored in the algorithm by this way. Each chromosome has the length corresponding to the number of reactions in the full mechanism. Each individual represents a kind of sub-mechanism. For initial population, all the chromosomes are set to the same and all genes assigned as the value of 1 represent the original mechanism. This approach is designed to avoid the loss of key reactions in the first generation. The iteration between generations seeks the optimal reaction subset under user-defined evaluation rules.

The evaluation function (objective or fitness function) is an essential part of this optimization technique. Different sub-mechanisms are selected as the resulting reduced mechanism under various evaluation function definitions. Three parts including the accuracy, size, and stiffness of the reduced mechanism are taken into consideration. The accuracy of sub-mechanism is evaluated by simulating combustion in a homogeneous reactor, where ignition delay time and steady-state temperature are selected as the properties of interest. Generally speaking, the fewer reactions in the mechanism are, the smaller the calculation cost is, yet not including slow converging mechanisms. That is why the CPU calculation time is also tracked to assess the stiffness of the sub-mechanism. According to Sikalo et al. [37], the ignition delay time error and the steady-state temperature error are defined as Equations (1) and (2) respectively due to the logarithmic expression steep convergence toward the optimal point. The mechanism size and cost evaluation function are defined as Equations (3) and (4). The global objective function is defined in Equation (5).

$$f_{\tau} = \ln \left(1 + \sigma \left| \frac{\tau_r - \tau_0}{\tau_0} \right| \right) \quad (1)$$

Where τ_r is IDT simulated by sub-mechanism, and τ_0 is IDT simulated by full-mechanism.

$$f_{Ts} = \ln \left(1 + \sigma \left| \frac{T_{s(r)} - T_{s(0)}}{T_{s(0)} - T_{in}} \right| \right) \quad (2)$$

Where $T_{s(r)}$ is steady-state temperature estimated by sub-mechanism, $T_{s(0)}$ is steady-state temperature estimated by full-mechanism and T_{in} is initial temperature.

$$f_N = \left[1 + \exp \left(\sigma \left(1 - \frac{N_r}{\lambda \times N_R} \right) \right) \right]^{-1} \quad (3)$$

Where N_r is reaction number of sub-mechanism, and N_R is reaction number of full-mechanism.

$$f_{CPU} = \left[1 + \exp \left(\sigma \left(1 - \frac{t_{CPU(r)}}{\lambda \times t_{CPU(0)}} \right) \right) \right]^{-1} \quad (4)$$

Where $t_{CPU(r)}$ is CPU cost time of sub-mechanism, and $t_{CPU(0)}$ is CPU cost time of full-mechanism.

$$f_{objection} = f_{\tau} + f_{Ts} + f_N + f_{CPU} \quad (5)$$

The performance in different aspects can be improved simultaneously and the weights of each component can be easily adjusted by the GA optimization method. For guaranteeing a good trade-off between concerned terms, the sharpening coefficient σ and shifting coefficient λ require special tuning. Fig. 3 compares the effects of factors on each evaluation function. Parameter values for the objection function terms are shown in Table 1.

2.3. Implementation details of mechanism optimization

In the application of mechanism reduction above, the sub-mechanism and the original mechanism are required to be as consistent as possible while predicting the heat release and ignition delay characteristics of ammonia/hydrogen. Apart from ignition delay characteristics, the flame propagation velocity of hydrogen mixture was also concerned here. In order to decrease the deviation between the calculations by the resulting reduced mechanism and measurements in experiments, the reduced mechanism was improved by tuning Arrhenius pre-exponential factor A based on the genetic algorithm.

The goal of the optimization is the A-factor of all reactions in reduced mechanism except for the hydrogen sub-mechanism and the pressure-dependent reactions, including 45 reactions. It is considered that hydrogen oxidation has been studied sufficiently and the current hydrogen mechanism is quite mature [20]. The pressure-dependent reactions are also unchanged for simplicity. The uncertainty factor is set as 10, which means the variation range of pre-exponential factors is $[0.1 \times A_i, 10 \times A_i]$, where A_i is the pre-exponential factor of i th reaction. It is noted that GA with binary encoding method may fail when handling multi-dimensional, high-precision numerical problems due to the excessive search space, called curse of dimensionality. Therefore, Elliott et al. [40] proposed a GA with real code or floating point (FP) genes to deal with the chemical kinetics optimization problems. Since there are dozens of reactions needing modification, the real encoding approach was adopted in this study. Similar to mechanism reduction for initial conditions, the starting point (generation 1) of iteration is the reduced resulting mechanism. The important GA parameters were exhibited in Table 2.

Except for ignition delay time error and calculational effort, Equations (6) and (7) were also introduced to constrain laminar flame speed (LFS) error and the amplitude variation of the reaction rates. The parameters in the objection function were shown in Table 1. Due to the uncertainty in experimental data and the approximation between measurements and predictions by mechanism, the linear expression was used for LFS accuracy criteria. Wang et al. [41] recommended that it should be kept an optimized mechanism faithful to the original mechanism, and his evaluation equation for Pre-A variation was referred to here. The objection function was defined as follows:

$$f_{LFS} = \sigma \left| \frac{u_l^{\exp} - u_l^{pre}}{u_l^{\exp}} \right| \quad (6)$$

Where $u_l^{\exp}$ is experimental LFS data, and u_l^{pre} is predicted LFS data

$$f_A = \left| \sigma \times \ln \left(\frac{A}{A_0} \right) \right| \quad (7)$$

Where A_0 is initial Arrhenius pre-exponential factor, and A is adjusted Arrhenius pre-exponential factor.

$$f_{objection} = f_{\tau} + f_{LFS} + f_A + f_{CPU} \quad (8)$$

3. Result

The reduction operation was based on the comprehensive chemistry mechanism of ammonia/hydrogen developed by Wang et al. [21] in 2021, which consisted of 34 species and 263 reactions and was verified widely against various experimental conditions. A wide range of

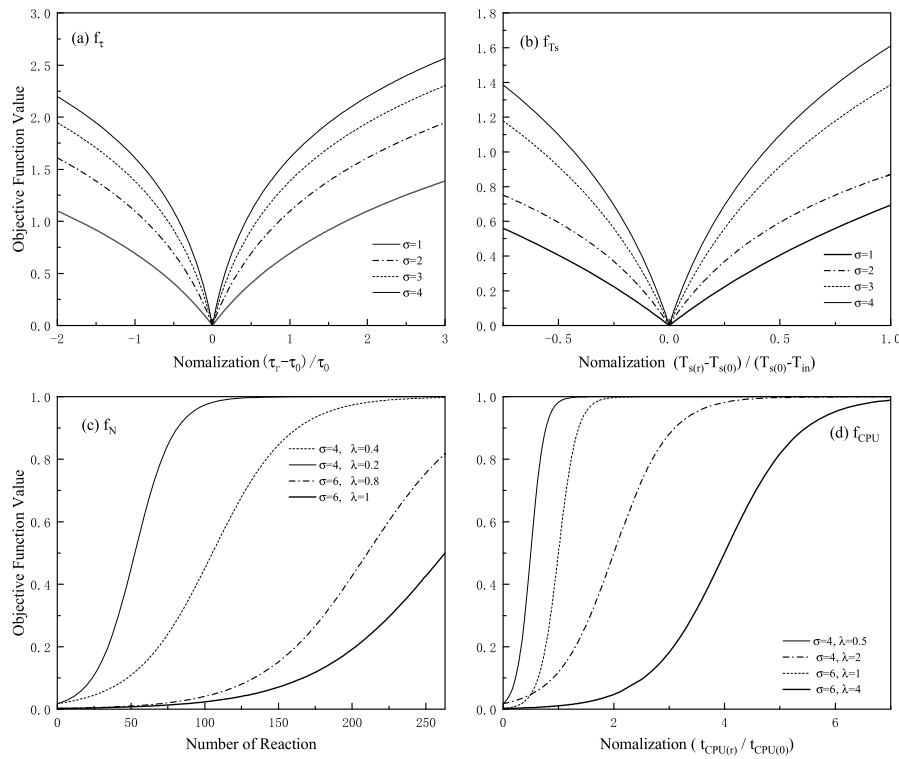


Fig. 3. Objective functions terms with different σ and λ .

Table 1

Coefficient values for the objection function terms.

Item		σ	λ
Reduction Mechanism	f_{τ}	1	
	f_{τ_s}	1	
	f_N	6	0.8
	f_{CPU}	6	4
Optimization Mechanism	f_{τ}	4	
	f_{LFS}	4	
	f_A	0.3	
	f_{CPU}	6	4

Table 2

Parameter values of genetic algorithm.

Item	Value for Reduction	Value for Optimization
Population size	60	30
Number of generations	1000	1000
Crossover rate	0.2	0.3
Mutation rate	0.03	0.06
Selection	Tournament selection	Tournament selection
Crossover	Single point	Single point
Mutation	Two directional	Two directional

conditions was contained as optimization targets and the details were shown in Table 3. The homogeneous mixture simulation in case 1–8 was employed when reducing the mechanism. For reaction rate modification, the free propagation flame model was also adopted to improve the performance of the reduced mechanism in case 9–12. All zero-dimensional and one-dimensional combustion numerical calculation in this study were finished with Cantera code.

3.1. Reduced resulting mechanism

As illustrated in Fig. 4, the overall objection function value dropped

Table 3

Overview of target environments for reduction and optimization.

Case No.	X[H ₂] in fuel	P (atm)	T (K)	phi
1	–	1	1540–1850	1
2	–	10	1430–1820	1
3	0.05	1	1500–1920	1
4	0.05	10	1450–1920	1
5	0.3	1	1180–1820	1
6	0.3	10	1180–1920	1
7	0.7	1	1000–1540	1
8	0.7	10	1030–1300	1
9	–	1	423	0.7–1.4
10	0.4	1–5	298	0.7–1.5
11	0–1.0	1	298	1
12	0.05	1	298–473	0.8–1.4

rapidly with iterations in the first 200 generations. The overall score disparity among the fittest, the worst, and the average of the population was not obvious during evolution, which showed that the direction of evolution was relatively consistent. The four objective function values were turned for balance as much as possible in our study. In the end, the overall values of these three reduced to around 0.04, which included the reactions number penalty items (around 0.014), therefore the final reduced mechanism and the full mechanism had similar capabilities in accurately predicting the ignition delay times and state-state temperature under investigation conditions. In the 1000th generation, the overall objection function values of the three showed no significant difference (0.03938, 0.03987, 0.03955, respectively). Based on this, it was considered that GA converged or almost converged and the benefits from continued calculation were very limited. The number of reactions in the best individual only experienced a small change after 500 generations. Fig. 5 gave more details about the genes of the fittest in each generation. In this article, the direction of gene mutation was set as two-way, unlike the previous studies [36,37]. On the one hand, the small number of genes (compared to the number of individuals in the

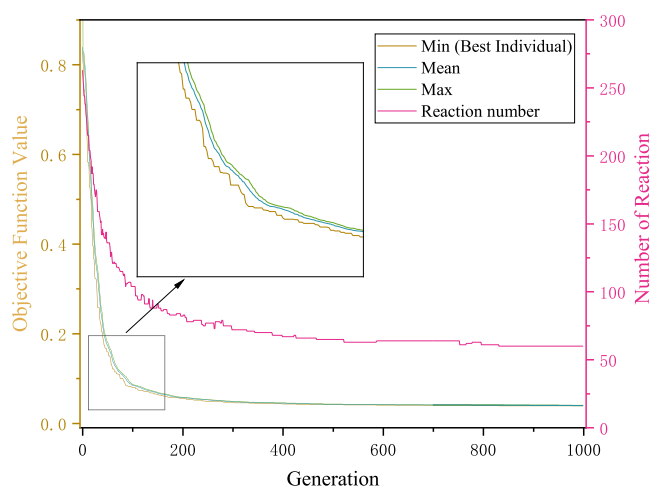


Fig. 4. Overall objection function value for mechanism reduction during evolution.

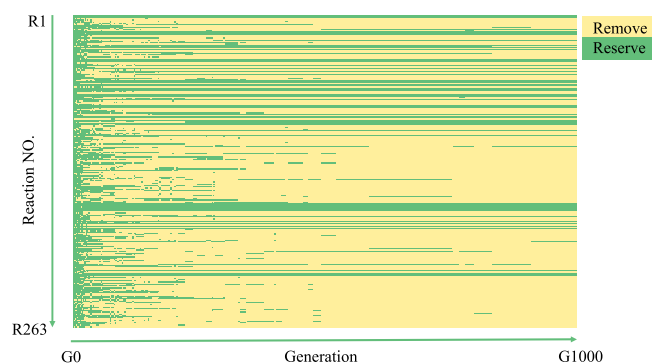


Fig. 5. Gene information of the best individual in every generation for mechanism reduction.

population) and the tiny mutation probability ensured the robustness in the process of GA reduction. On the other hand, it was aimed to be more tolerant of individuals that lost important reactions and search solutions in a larger domain. The final reduced mechanism contained 29 species (including CO, CO₂, CH₄, C₂H₆ only defined as three-body colliders, Ar and He) and 63 reactions, called Mechanism 1 in this study.

The sensitivity to ignition delay times of each reaction in the full mechanism was detected and the sensitivity coefficient was normalized under each operating condition. The environments of the sensitivity analysis were included in the optimization targets. The top 20 most ignition-sensitive reactions were plotted in Fig. 6. After the addition of hydrogen to ammonia, the reaction $H + O_2 = O + OH$ became the most essential reaction to promote ignition [42]. From Fig. 6, it was found that the GA automatically selected most of the key reactions for ignition prediction. It indicated that the GA reduction technique was reliable in this study, even without chemistry. To understand the NO production and consumption path in Mechanism 1, the analysis of production rate of ammonia/hydrogen/air mixture at high temperature, atmospheric pressure, $[NH_3]: X[H_2] = 0.8:0.2$ and $\phi = 1$, were conducted. From Fig. 7, the NO was mainly formed through reaction of $HNO + H = NO + H_2$, followed by reaction $N + OH = NO + H$ in this condition for both full mechanism and Mechanism 1. Besides, the reaction $NH_2 + NO = NNH + OH$ and $NH_2 + NO = N_2 + H_2O$ were considered as most important for NO reduction [21,43] and also reserved in Mechanism 1. The reactions contributing significantly to the N₂O concentration of full mechanism were shown in Fig. 7(c), and it can be seen that the reactions $NH + NO = N_2O + H$, $N_2O + H = N_2 + OH$ and $N_2O(+M) = N_2 + O(+M)$ dominated overwhelmingly. All those three reactions were retained in the simplified mechanism proposed in this paper.

3.2. Optimized resulting mechanism

Since the genetic algorithm in the optimization procedure was also a single-objective and gene real-coded one, which made it possible to inherit the previous program framework used for mechanism reduction and save the labor cost of code development. The target of the GA optimizer in this paper was the pre-exponential factors of 46 reactions. Although the search range was discretized, the occupation of time and calculation resources were still large if too many parameters would be adjusted. It was suggested that the reactions should be conditionally selected when optimizing a large-scale mechanism.

Fig. 8 showed the detailed information of the population and the fitness in every generation. It could be seen that the population evolved rapidly in the first 200 generations to achieve greater fitness scores. From the 200th generation to the 600th generation, the mean objective function value of the population decreased slowly, and the gap between the average and min was further reduced. The objective function values of the best individual and population average in the 1000th generation were 3.07786, 3.05898 respectively and both were almost unchanged from 700th to 1000th. Specifically, the objective function value of the fitness decreased in a step-like manner. This was due to the

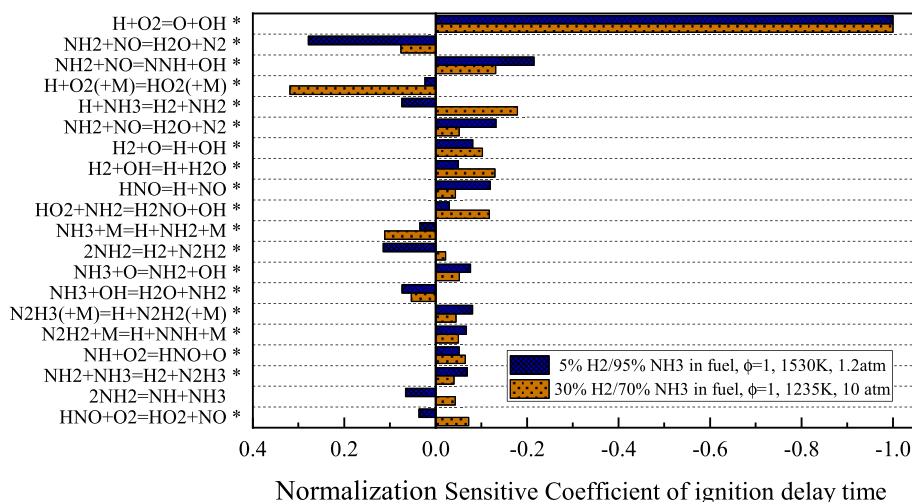


Fig. 6. Normalized sensitivity of the ignition delay times for various conditions (* represented this reaction was retained in Mechanism 1).

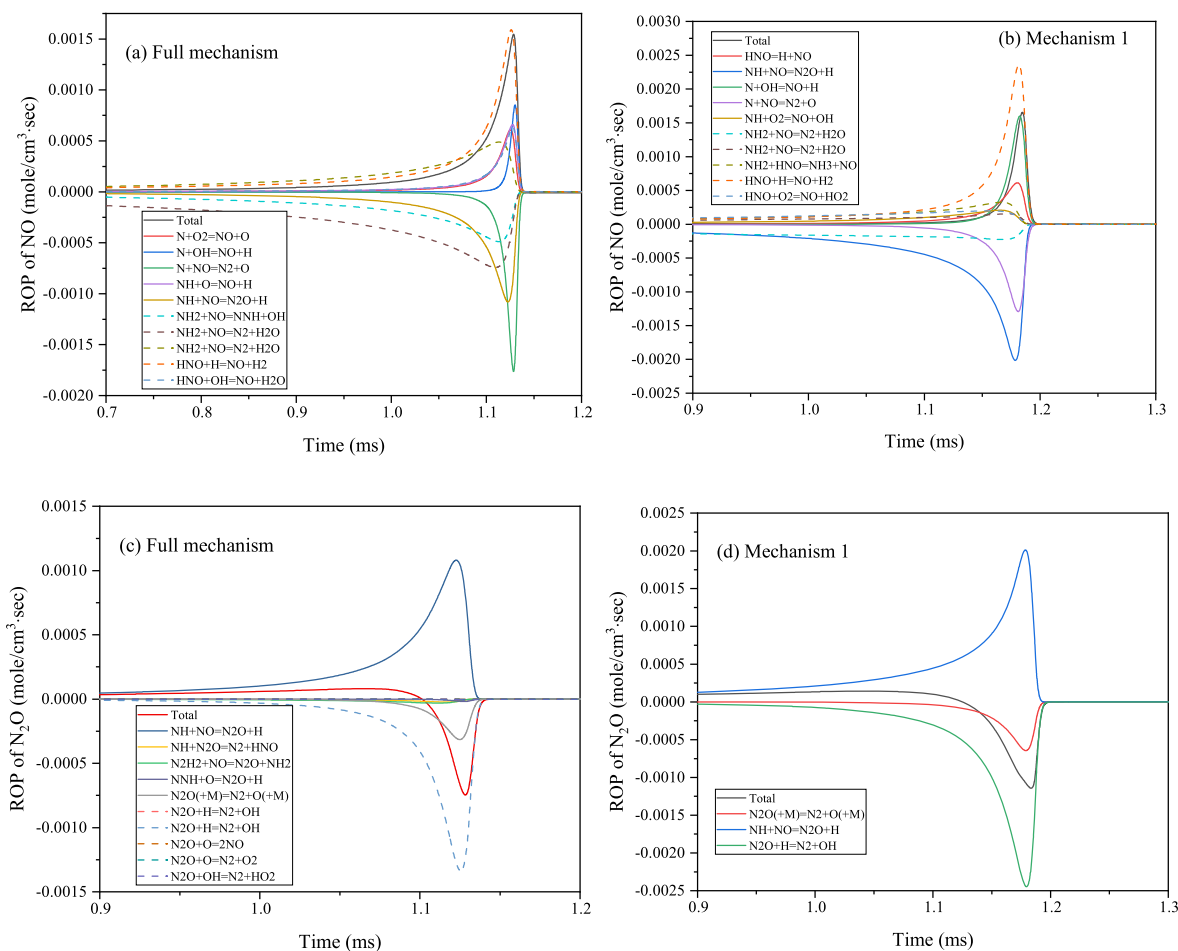


Fig. 7. Analysis of production rate using reduced mechanism (Mechanism 1) and detailed mechanism for NO and N₂O under 1200 K, $\phi = 1$, 80%NH₃/20%H₂ as fuel.

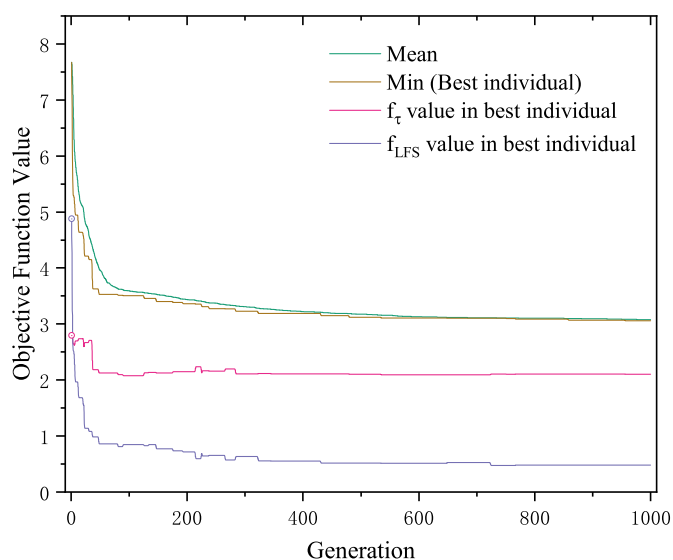


Fig. 8. Objection function value for mechanism optimization during evolution.

discontinuous optimization space and random gene mutation. The emergence of the new best individual can be attributed to beneficial mutations or crossovers in the genes of the parent. In addition, it can be concluded from Fig. 8 that compared with the initially reduced mechanism, the advantages of optimized reduced mechanism reflected in the

prediction of laminar flame velocity, while the improvement of the ignition delay time term was relatively small. The reason for that would be discussed in the mechanism verification part. Both reduced and optimized mechanisms were provided as Cantera YAML format in supporting material.

4. Mechanism verification

The detailed mechanism was processed by the GA, and a compact mechanism set (Mechanism 1) was extracted. After that, Mechanism 1 was optimized to reduce disagreement between model predications and experiment measurements of IDTs and LFS obtained in literature. Although the global fitness of the reduced mechanism and the optimized mechanism was satisfactory as described above, the mechanisms still needed to be further verified against experimental data under a wide range of operation conditions.

4.1. Verification of ignition delay time

In CI engines, fuel's ignition behavior was one of the most interested characters during combustion progress. The ignition characteristics of fuels were dominantly governed by chemical kinetics and the experimental data of IDTs were often applied to test the chemical mechanism. As a consequence, the IDTs data from the shock tube were utilized to assess the reduced mechanism in this section.

Chen et al. [42] measured the IDTs of stoichiometric ammonia/hydrogen blends in a shock tube. The experimental range covered the temperatures from 1020 K to 1945 K, pressures from 1 atm to 10 atm,

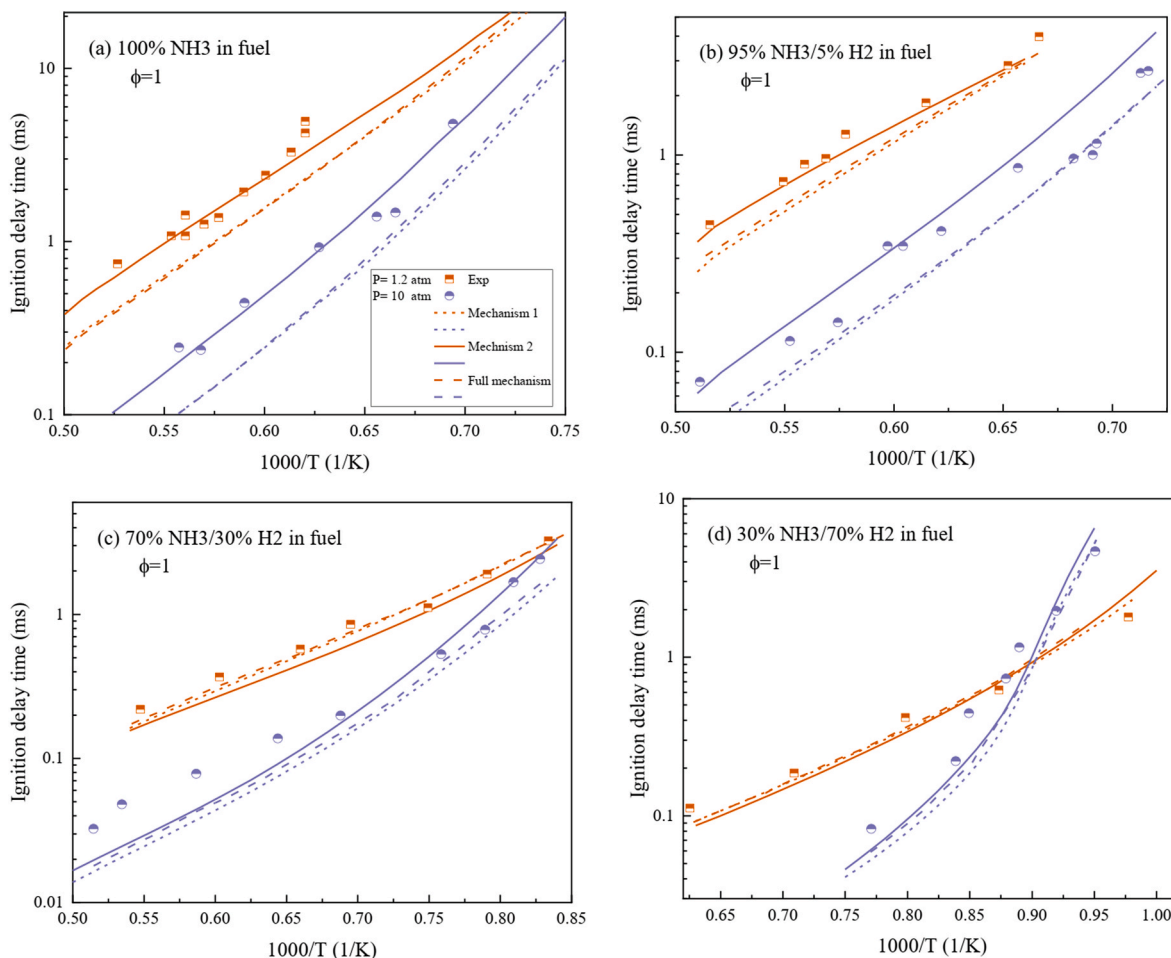


Fig. 9. Measured (symbols) [42] and calculated (lines) IDTs by optimized mechanism, initial mechanism, and reduced mechanism.

and hydrogen fractions from 0% to 70%. As a comparison, the predicted results of the original mechanism were also plotted in Fig. 9. The spontaneous combustion temperature of ammonia was high. The ignition delay times became shorter with the mixing of hydrogen. As in Fig. 9 (d), when the proportion of hydrogen was too large, the IDTs of the ammonia/hydrogen mixture was dominated by the hydrogen chemistry [42] and negative pressure dependence characteristic of the IDTs occurred. It can be found that the simulation IDTs estimated by reduced mechanism and the original mechanism showed no obvious distinction in Fig. 9. This was due to that the ability to forecast ignition behavior of the subset mechanism was evaluated with full mechanism as the standard under the guidance of the penalty function. Comparison between various calculated values and experimental IDTs data demonstrated that the Mechanism 2 strived to balance accuracy under broad operation conditions, and performed better in the overall prediction. In addition, the calculational line by the original mechanism and Mechanism 1 deviated greatly from the measurements in Fig. 9 (a), while the performance of Mechanism 2 was satisfactory.

Fig. 10 compared the predicted and measured ignition delay times of pure ammonia at different temperatures, equivalence ratios, and pressures. It can be seen from Fig. 10 (a)–(c) that the variation tendency of the IDTs with temperature and pressure forecasted by Mechanism 2 was consistent with the detailed mechanism. Additionally, the difference between the experimental data and the IDTs predicted by the optimization mechanism was obvious. This was due to GA optimization being more inclined to find the balanced and optimal solution under the global optimization environments. The conformity illustrated the reliability of the mechanism to simulate the ignition delay behavior of the ammonia/

hydrogen mixture under the investigation conditions. The predicted ignition delay times of ammonia at high temperatures was also verified as shown in Fig. S1. The ability of the proposed ammonia/hydrogen reduced mechanism to predict the combustion of dilute mixture was illustrated in Fig. S2. Currently, the ability of the detailed ammonia/hydrogen mechanism to predict the combustion at the condition of low-temperature and dilute mixture was still not ideal [9], and the performance of our proposed simplified mechanism was slightly improved.

4.2. Verification of laminar flame speed

The laminar flame speed of combustible mixture is a fundamental global parameter and could be used to validate the feasibility of reaction mechanisms. The ammonia has low flame propagation speed and high ignition energy, which brings difficulties into its application in SI engines. The mixing of hydrogen can promote the combustion speed of the ammonia/hydrogen/air mixture. This section verified the reduced and optimized mechanism against experimental laminar flame speed. The problem was solved with a mixture-averaged transport model.

The values of laminar flame velocity predicted by the full mechanism, Mechanism 1 and Mechanism 2 were compared with the experimental data in Fig. 11. It can be seen that the laminar flame speed of the ammonia/hydrogen mixture predicted by Mechanism 1 was not credible, and the performance of Mechanism 1 was greatly improved by adjusting the A-factors. More comparisons between Mechanism 2 and the initial mechanism were shown in Fig. 12. Under the experimental conditions described in Fig. 12 (a), the full mechanism overestimated the laminar flame velocities, and Mechanism 2 estimated more reliable

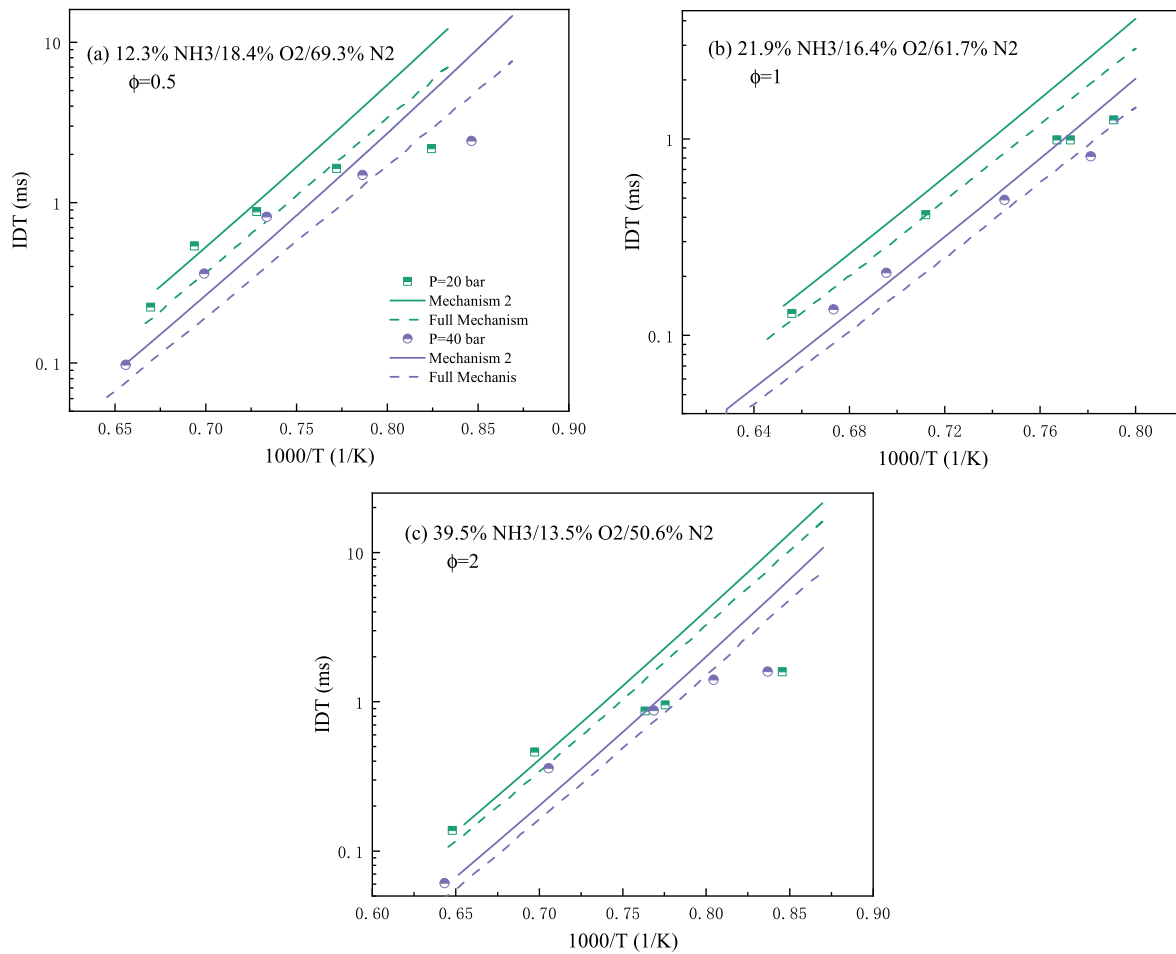


Fig. 10. Ignition delay times of NH₃/air [44] and NH₃/O₂/Ar mixture [17] measured in a shock tube at various equivalence ratios.

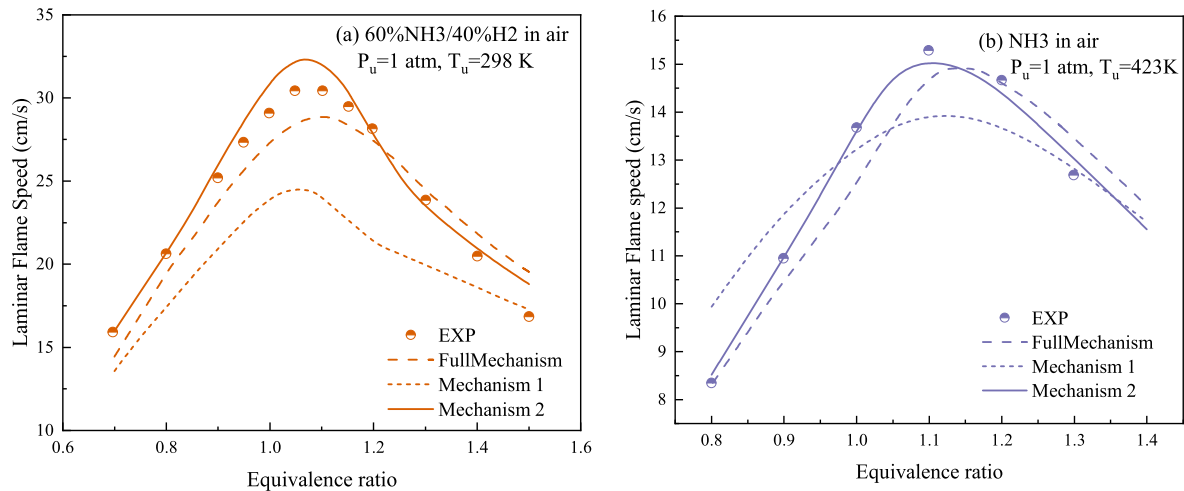


Fig. 11. Measured [45,46] (symbols) and calculated (lines) LFSs by optimization mechanism, full mechanism and reduction mechanism.

data at the condition of the fuel-rich mixture.

The prediction performance of the full mechanism, Mechanism 1 and Mechanism 2 under different proportions of hydrogen, initial pressure from 1 to 5 atm was depicted in Fig. 13. The calculated values from the optimization mechanism agreed slightly better with experimental data at $p = 1$ atm compared with the full mechanism. Mechanism 2 estimated a smaller value than the full mechanism in low mole fraction regions of H₂ (<0.3) under the initial pressures of 3 and 5 atm. At the high

proportion of hydrogen operation conditions, the NH₃/H₂/air mixture was governed by H₂ chemistry. From Fig. 13 (c), the predicted data by Mechanism 1 and Mechanism 2 were similar and both higher than those of the full mechanism. The experimental data of H₂/O₂/He laminar flame speed [47] at $P = 5$ atm and $\phi = 1$ provided some confidence about the reliability of hydrogen sub-mechanism in the reduced mechanism.

As seen from Fig. 14, both Mechanism 2 and the full mechanism can

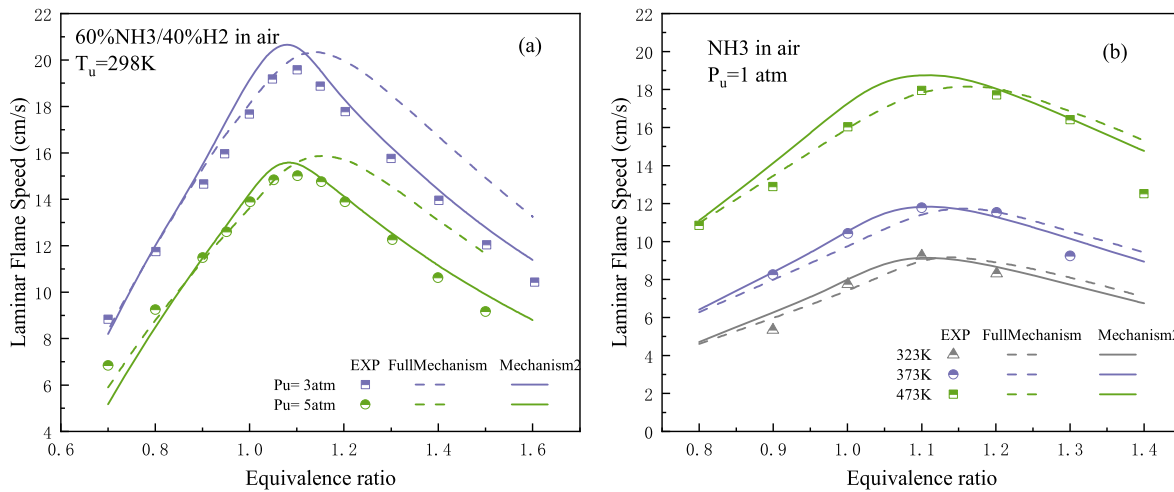


Fig. 12. Measured [45,46] (symbols) and calculated (lines) LFSs by optimized mechanism, full mechanism, and reduced mechanism.

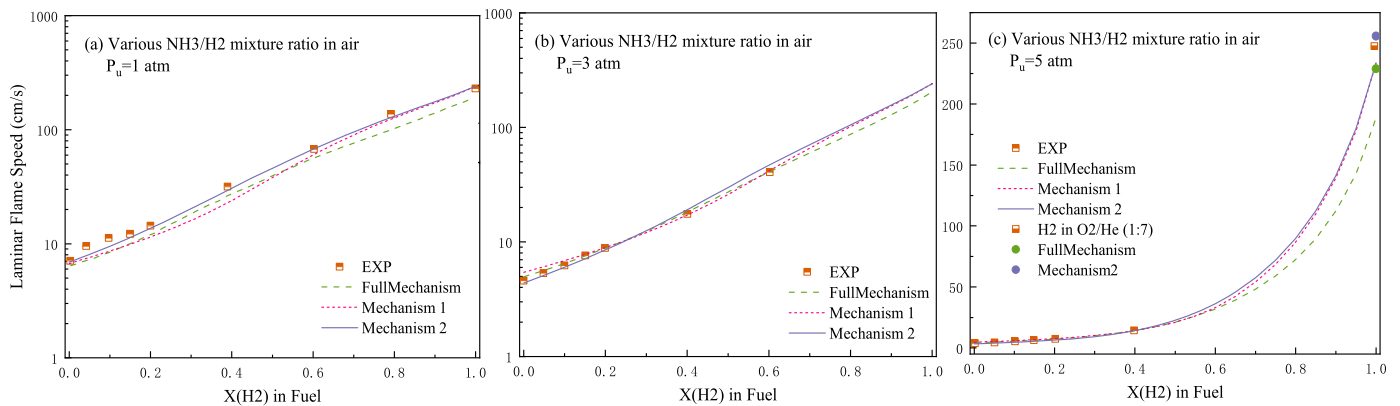


Fig. 13. Laminar flame speed of NH₃/H₂/air mixtures under atmospheric and pressurized conditions ($\phi = 1$), initial mixture temperature of 298 K, experimental data from Refs. [47,48].

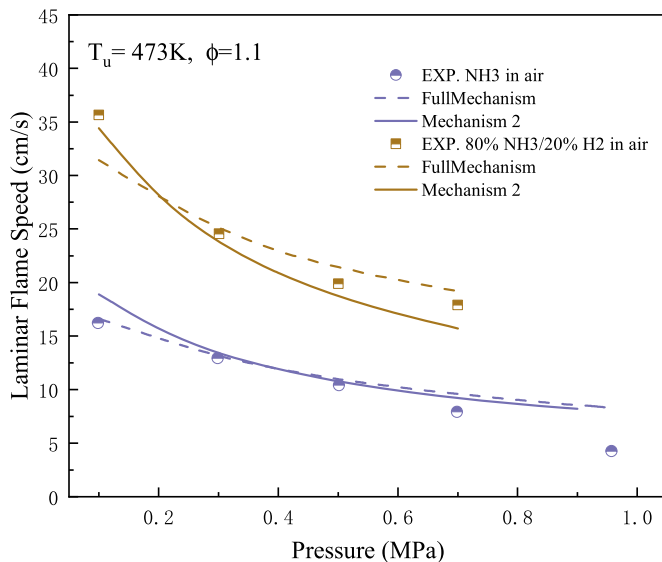


Fig. 14. Comparison between measured [49] (points) and calculated (lines) laminar flame speeds of NH₃/H₂ mixtures with different pressures.

capture the variation tendency of laminar flame speed with pressure. Besides, Mechanism 2 was verified under a wide range of experimental conditions. As shown in Fig. 15, the verification range covered temperature of 298–473 K, equivalence ratio of 0.8–1.4, and hydrogen ratio of 0.05–0.6. Accuracy of hydrogen sub-mechanism was slightly reduced due to the simplification of the hydrogen sub-mechanism in full mechanism, which caused a maximum 10% estimation error at 473 K in Fig. 15(c). However, the performance of Mechanism 2 remained satisfactory in general under wide range of operating conditions compared with original mechanism. The good agreement between the experimental data and the prediction proved that the optimized mechanism was credible to reproduce the laminar flame speed characteristics of the ammonia/hydrogen mixture.

As shown in Fig. S3, the laminar flame speed verification was extended to high temperature and high pressure and the mechanism still performed well. As displayed in Fig. S4, the pressure dependence of the laminar flame speed became more pronounced when the hydrogen doping ratio increased, and our mechanism exhibited a similar pattern. Besides, the present model was also reliable to predicted the NH₃/H₂/N₂/air mixture flame, as illustrated in Fig. S5. And the model of Stagni et al. [20]'s and experimental data of Ji et al. [50] was adopted to illustrated present mechanisms.

4.3. Verification of species concentration

The main species concentration profiles from the jet-stirred reactor (JSR) and premixed flame experiments of NH₃ or NH₃/H₂ mixture were

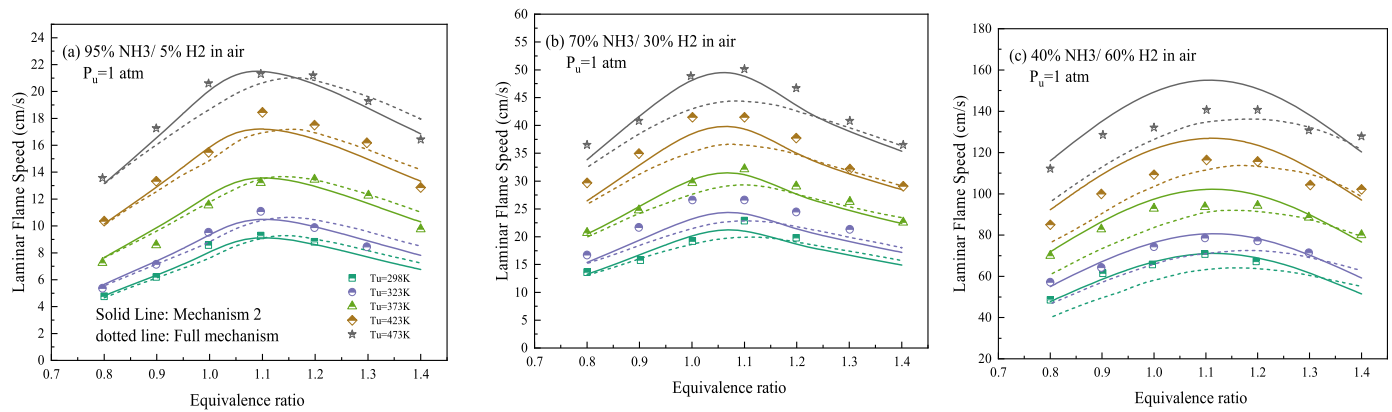


Fig. 15. Comparison of measured laminar flame speed [45] and calculated data by Mechanism 2 of NH₃/H₂ mixtures.

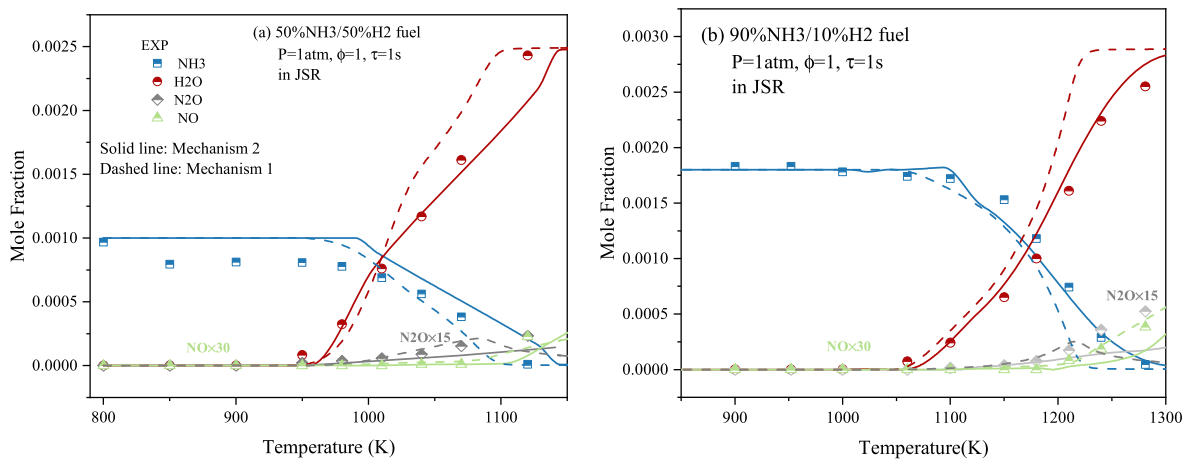


Fig. 16. Measured [21] and simulated mole fractions of NH₃, H₂O, N₂O, and NO in JSR oxidation of NH₃/H₂ mixtures with various H₂ content.

utilized to evaluate Mechanism 2 in this section. The stirred Reactor model in Cantera was used to simulate a perfectly stirred reactor. The Burner Flame model in Cantera was implemented to calculate premixed ammonia/hydrogen flame structure. As shown in Fig. 16, the concentration evolution of NH₃ and H₂O estimated by Mechanism 2 was in good agreement with the measurements. The Mechanism 1 exhibits a stronger reactivity and a lower conversion temperature of NH₃ relative

to Mechanism 2. Although the predicted concentrations of NO and N₂O were lower than the experimental values, the history trends were basically identical.

Fig. 17 compared the premixed flame structures simulated by the original full mechanism and Mechanism 2. It was observed that the NO concentration in the post-flame zone still couldn't be ignored. The widths of the reaction zone predicted by Mechanism 2 and the initial

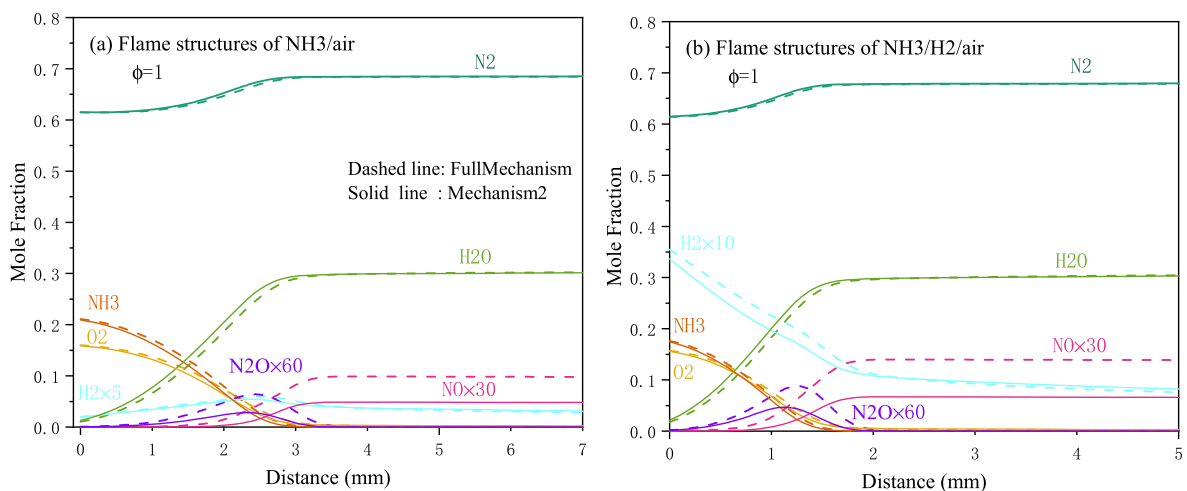


Fig. 17. Premixed flame structures at $\phi = 1$ simulated by different models, (a) pure NH₃, unburned temperature of 350 K and inlet velocity of 7.8 cm/s, (b) X[NH₃]: X[H₂] = 0.8:0.2, unburned temperature of 350 K and inlet velocity of 13.2 cm/s.

mechanism were not significantly different. The estimated NO and N₂O species concentrations by Mechanism 2 were approximately 1/2 of those from the original mechanism. However, the main channels of N₂O generation and consumption are remained in the simplified mechanism in this paper. More verification of NO distribution in premixed flame can be found in Appendix. According to Fig. S6, it was found that the calculated NO concentration of Mechanism 1 and Mechanism 2 were still lower than that of the full mechanism, however prediction result were closer to the experimental measurements for stoichiometric and rich NH₃/N₂/air mixture. In order to test the reaction paths related to NO consumption, comparisons were illustrated in Fig. 18 between experimental measurements [51] and models calculations of NO/N₂/H₃/Ar premixed flame. The experimental temperature data were employed as input data for the simulation. Generally speaking, Mechanism 2 and full mechanism gave very similar predicted species profiles and the simulation basically agreed with the experimental data.

4.4. Verification of the combustion in HCCI engine

Different from the conventional CI combustion mode, the HCCI combustion process was mainly controlled by chemical reaction kinetics [52] as the homogeneous premixed mixture in-cylinder. The reduced mechanism was validated against the in-cylinder pressure data of the HCCI engine contributed by Pochet et al. [11]. The engine specifications and operating conditions were listed in Table 4. The calculation was done in commercial 3D CFD software and the RNG k- ϵ model was chosen for turbulence simulation. Due to the symmetry of the combustion chamber, a 60° grid was established to reduce computational cost. The 2.0 mm base grid and mesh adaptive refinement are employed. The number of calculation grids was more than 400,000 when the engine was at the top dead center and was displayed in Fig. 19. The simulation began at the intake valve closing (−127°CA BTDC) and ended at the exhaust valve opening (127°CA ATDC).

Fig. 20 compared the measured and predicted in-cylinder pressures at various hydrogen blend ratios. It can be observed that the peak in-cylinder pressure simulated by the initial mechanism, Mechanism 1 and Mechanism 2 exhibited no obvious difference and the relative error between predicted and measured data was within 5%. The estimated ignition delay time by the three mechanisms was significantly different, and the simulated pressure curve by Mechanism 2 was closer to the experimental profile. This was due to the reduced mechanism being optimized to accurately forecast the ignition delay time under interesting conditions.

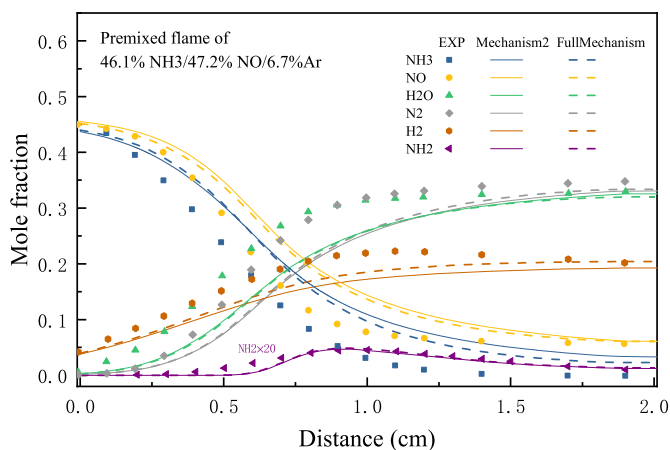


Fig. 18. Premixed flame structures of NH₃/NO/Ar mixture, $\phi = 1.46$ and inlet velocity of 60 cm/s, experimental data from Ref. [51].

Table 4

Engine specifications and operating conditions.

Parameter	Value
Displacement(cm ³)	463
Bore × Stroke (mm)	86 × 75
Compression ratio	22
Speed(rpm)	1500
equivalence ratio	0.22
X[NH ₃]:X[H ₂]	0.37:0.63/0.57:0.43
Intake pressure(bar)	1

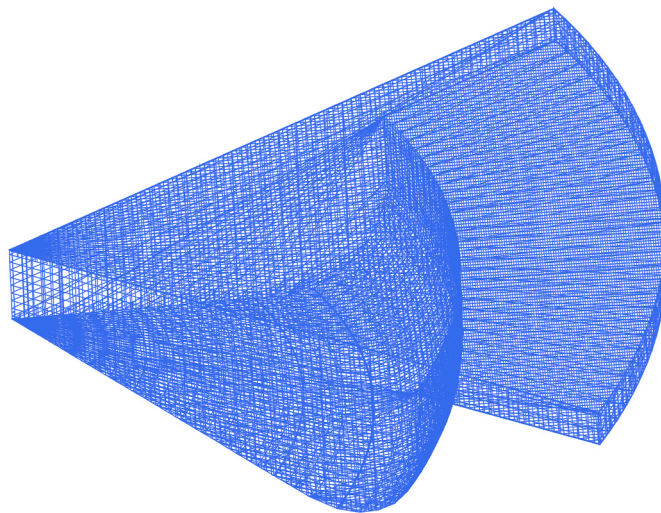


Fig. 19. Computational grid in engine simulation.

5. Conclusion

The reduced mechanism and optimized mechanism for ammonia/hydrogen mixture were both developed through the single-objective genetic algorithm. The reduced mechanism included 29 reactants (including CO, CO₂, CH₄, C₂H₆ only defined as three-body colliders, inert gas Ar and He), and 63 element reactions.

During the mechanism reduction progress, not only the IDT and steady-state temperatures were included as evaluation terms to keep high fidelity with the initial mechanism, but also the calculational cost time by the reduced mechanism was restricted to avoid stiff mechanism. The sensitivity analysis of ignition delay time illustrated the most important reactions to ignition which were retained in the reduced mechanism. This indicated that GA was a feasible methodology for reducing detailed mechanisms, which had flexible and adjustable evaluation rules and didn't require the user to understand the combustion chemistry.

The reduced mechanism was optimized based on GA as the baseline of fundamental experimental data. The predicted IDTs and LFS, pre-A variation amplitude, and calculation time were considered comprehensively for optimization. The overall score showed that the optimized mechanism performed much better on forecasting LFS. In addition, mechanism verification and comparison were carried out in shock tube, laminar flame speed, JSR, premixed flame, and HCCI engine environment. The consistency between the predictions and the measurements demonstrated the reliability of the reduced mechanism and optimized mechanism.

Author contribution

Xing Liu: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. Ying Wang: Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

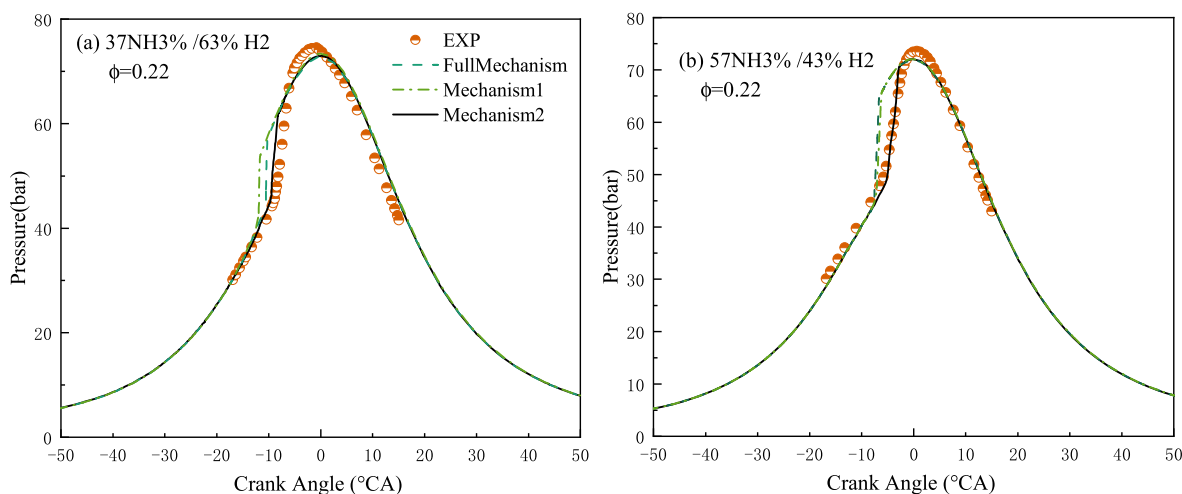


Fig. 20. Comparisons of in-cylinder pressures between simulation and experiment [11].

Yuanqi Bai: Software, Investigation. Wenxu Yang: Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This work was supported by National Natural Science Foundation of China (51776162) and Overall Layout Design and Research of High Efficiency Aviation Kerosene Rotor Engine Project (2020-JCJQ-ZD-116-11-01).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2023.126927>.

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